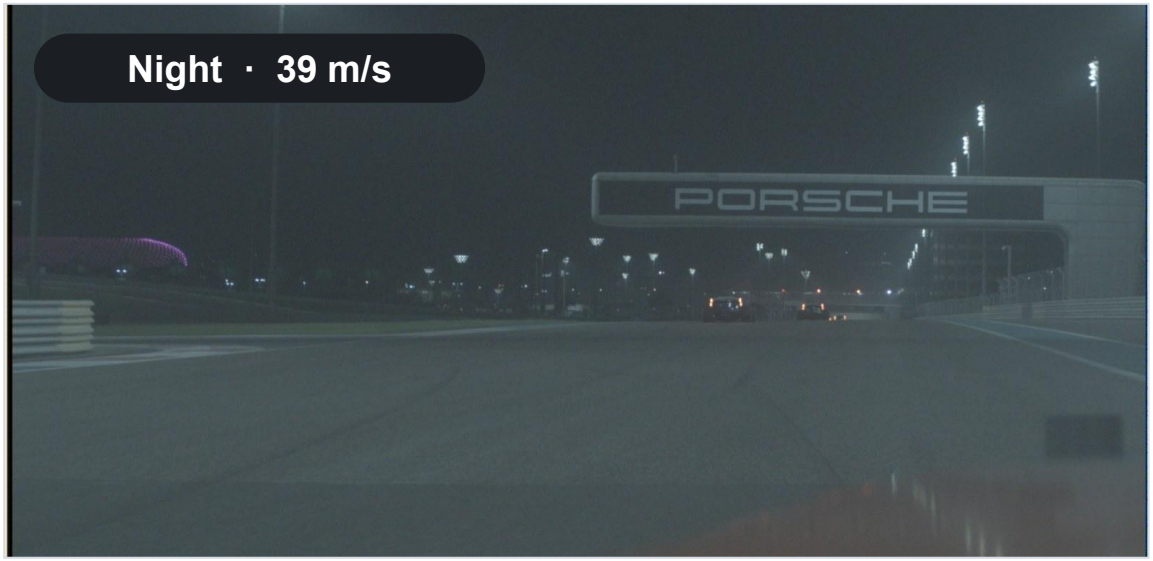
Team name: fill in

Where is the car?

Cameras only. No GNSS, no LiDAR, no peeking at future frames.



Training: 1194 s of night driving



Hidden test: may be faster, in different light

10 Hz one pose for every camera frame

4 m driven between two frames at 40 m/s

< 1 m sideways accuracy needed to hold a racing line

OUR IDEA

Recognise the place. Track the motion. Fuse both.

WHAT THE CAMERAS SEE

4 cameras
Front and side views, cropped to the track

→

Place network
ResNet-18 turns each view into a 256-number fingerprint

→

Map lookup
Finds the most similar training views near our estimate

→

HOW THE CAR MOVED

Car telemetry
Speed and turn rate, as the rules allow

→

Dead reckoning
Adds up distance and turning since the last frame

→

Cross-check
Motion learned from camera pairs is the backup

→

Kalman filter

Predicts with the motion, then corrects with the map match.

Each is weighted by how much it is trusted.

→

Pose
x, y, heading, 10 times a second

In plain words: Fingerprint
A list of 256 numbers describing a view. Similar views give similar numbers.

In plain words: Map
Every training frame stored with its fingerprint and its known position, night and synthetic daylight.

In plain words: Kalman filter
A standard way to blend a prediction with a measurement, trusting each by its confidence.

HOW IT HANDLES THE HARD PARTS

Each known difficulty gets a specific answer

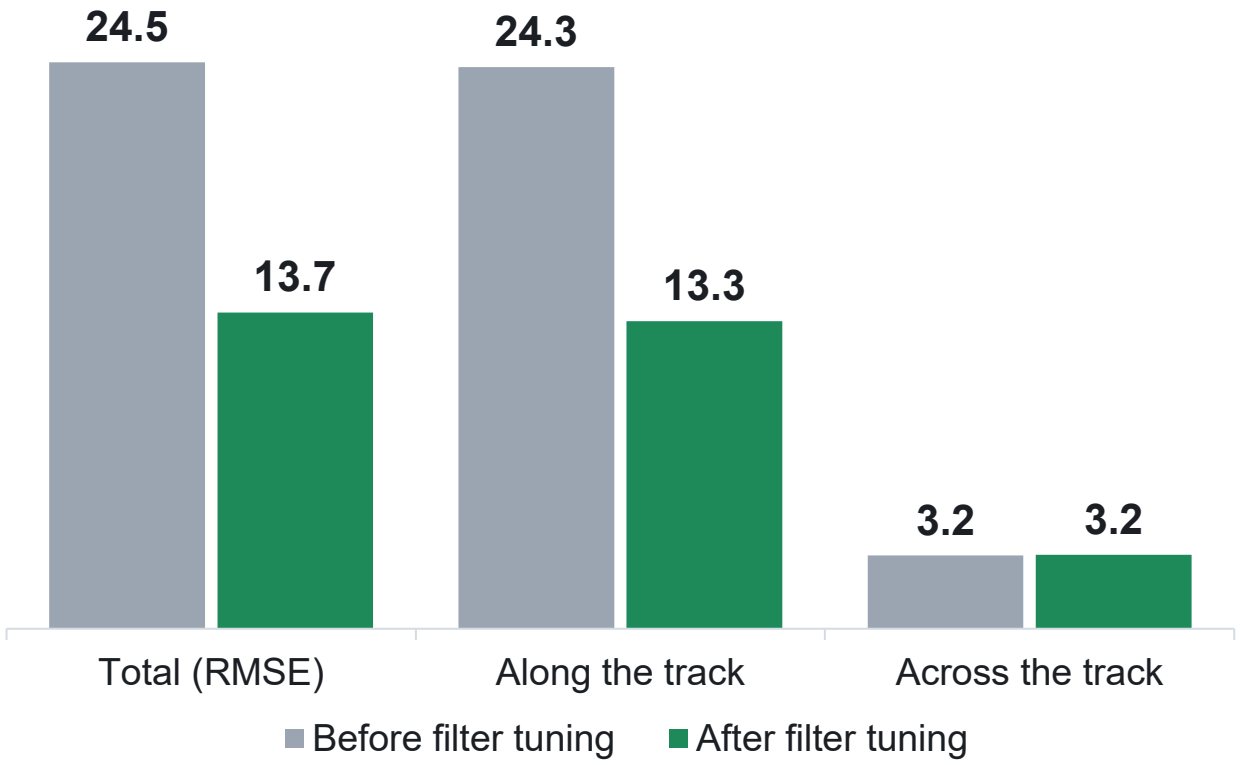
THE DIFFICULTY	OUR ANSWER	WHEN IT RUNS
Trained at night, tested in unknown light	**Synthetic daylight for training** A CycleGAN turns night frames into daylight look-alikes; the map stores both.	training only
Look-alike straights	**Learn places, then trust by similarity** Training pairs show one place on different laps. Weak matches count less; only nearby matches are averaged.	training + filter
4 m between frames, blur, cameras out of sync	**Strictly causal camera sync** Each camera supplies its newest image stamped no later than now. A learned gate weights the four views.	every frame
Drift, and getting lost	**Motion from the car, global re-lock** Speed and turn rate drive the prediction. Three confident matches elsewhere trigger a re-lock.	every frame

Training-only parts add no cost at run time. These are design choices; the next slide shows the one measured result we have.

EVIDENCE

Measure first: a filter fix cut the error by 44%

Position error in metres (lower is better)



Held-out validation, earlier pipeline version (git tags rmse-24.508 and rmse-13.688).
Sources: README.md, brainstorming_and_diagnostics.md.

−44% position error, with no retraining: two filter trust settings changed

98%	of the squared error was along the track
9.0 to 1.5 m	median error
12.8% to 32.8%	of frames within 1 m of the truth

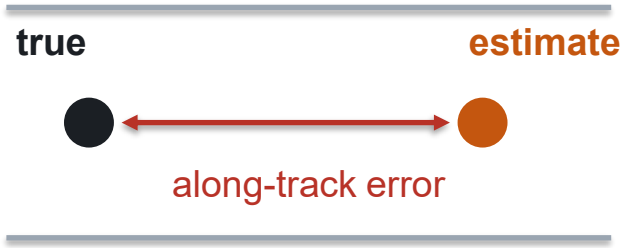
Final pipeline: fill in

Validation position RMSE	**fill in**
Across-track (lateral) RMSE	**fill in**
Time per frame	**fill in**
Hardware	**fill in**

FAILURE CASES

Where it still fails, and why

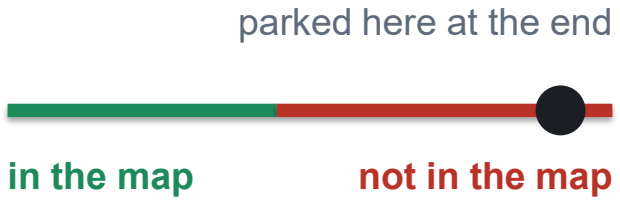
Look-alike straights



Along-track error dominated: 98% of the early squared error. After tuning, 5% of frames are still more than 20 m off.

Response: trust the map match more. Next: match short sequences of views.

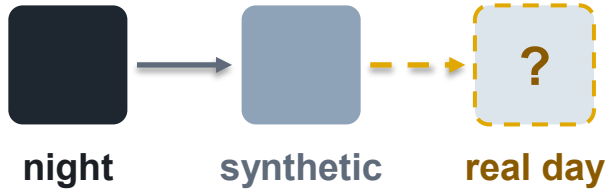
Places not on the map



An earlier validation run ended parked on an unmapped spot: 88% of its squared error came from the last 9 seconds.

Response: similarity check, hold-still rule, global re-lock. Not yet re-measured.

Unseen daylight



Synthetic daylight is generated, not measured. The daylight sample has no ground-truth poses to score against.

Response: lighting augmentation and night-to-day data; check speed against recorded telemetry.

Sources: README.md and brainstorming log (98%, 20 m); chats/astra.md debugging log (88%, 9 s); problem statement (daylight test).

LESSONS

Lessons learned and what comes next

1. **Diagnose before you train**
   An offline tool that may use ground truth showed the fault was the filter, not the network.

2. **Calibrated trust beats a bigger model**
   Two re-weighted settings gave minus 44%, with no retraining.

3. **Use every signal the rules allow**
   The car's speed and turn rate do not depend on light.

NEXT

Short sequences of views
instead of one frame at a time

↓

Geometric refinement
once the place is known: the path to sub-metre

↓

Pick the best checkpoint
by trajectory error, not the last epoch

Declared pre-trained model: ImageNet ResNet-18 (torchvision). Unlabelled daylight-sample frames train the night-to-day translator.
Code, README and requirements.txt are in the repository.

Repo link: fill in